Chapter 14. Bott periodicity for the unitary group: the Grassmannian of shortest paths

Milnor, "Morse Theory", §23, rewritten for a first-year graduate student who wants to know what the theorem is for. Every proof is complete, except that, as in Milnor, two facts of homotopy theory are quoted: coset spaces of Lie groups are fibre bundles, and fibre bundles have exact homotopy sequences. The homotopy groups of spheres that are needed are taken from the companion volume. All of this is stated precisely in §3.

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0. Why is magnetic charge an integer, and what happens in higher dimensions?

In 1931 Dirac showed that if a magnetic monopole of charge g exists, then every electric charge e satisfies $eg = \frac{1}{2}N\hbar c$ (Gaussian units) for an integer N. In the language of Wu and Yang (1975), the quantum states of a charged particle on a sphere around the monopole are the sections of a complex line bundle over S^2. Over the northern and over the southern hemisphere the bundle is trivial; it is determined by how the two trivializations are glued along the equator, that is, by a map from the equator S^1 to the group $U(1)$ of phases. The integer N is the winding number of this map, an element of $\pi_1 U(1) = \mathbb{Z}$.

Ask the same question over higher spheres. A bundle of states with N internal components over S^n is glued from two trivial pieces by a map $S^{n-1} \to U(N)$, so such bundles are classified by $\pi_{n-1}U(N)$. Over S^4 these are the Yang–Mills instantons, labelled by an element of $\pi_3 SU(2) = \mathbb{Z}$ (Belavin, Polyakov, Schwarz and Tyupkin, 1975). Over S^3 there is nothing: every bundle is trivial. What happens over S^{100}?

Bott's answer: once $N \geq n/2$, the group $\pi_{n-1}U(N)$ no longer depends on N, and it is $\mathbb{Z}$ for n even and 0 for n odd. The twisting alternates forever between "an integer's worth" and "none". This chapter proves it by Milnor's route. On the special unitary group $SU(2m)$ the shortest paths from I to $-I$ form the complex Grassmannian $G_m(\mathbb{C}^{2m})$, and every other geodesic from I to $-I$ has index at least $2m+2$. By the theorem of Chapter 13 (Milnor's Theorem 22.1), the Grassmannian then captures the homotopy groups of the path space up to dimension $2m$, and a little homotopy theory turns this into $\pi_{i-1}U \cong \pi_{i+1}U$. In §5 we return to monopoles and instantons and list the complex vector bundles over spheres of dimension up to eight.

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1. The unitary group, its exponential and its metric (Milnor pp. 124–126)

Let $M_n(\mathbb{C})$ be the space of complex $n \times n$ matrices, a real vector space of dimension $2n^2$ with the Euclidean inner product

$$\langle X, Y \rangle = \operatorname{Re}\operatorname{tr}(XY^*) = \operatorname{Re}\sum_{i,j} x_{ij}\bar{y}_{ij}, \qquad |X| = \langle X, X\rangle^{1/2},$$

where Y^* is the conjugate transpose. Three properties are used throughout: $|XY| \leq |X||Y|$ (Cauchy–Schwarz entry by entry), $|X^*| = |X|$, and $|SXT| = |X|$ for S, T unitary (because $\operatorname{tr}(SXTT^*X^*S^*) = \operatorname{tr}(XX^*)$).

Definition. The *unitary group* is $U(n) = \{S \in M_n(\mathbb{C}) : SS^* = I\}$, the group of linear maps of $\mathbb{C}^n$ preserving the Hermitian inner product. For $A \in M_n(\mathbb{C})$, $\exp A = I + A + A^2/2! + A^3/3! + \cdots$.

Since $|A^k| \le |A|^k$, the series and all its term-wise derivatives converge uniformly on bounded sets, so exp is smooth.

Lemma 1.1 (Milnor's (1)–(4)). (1) $\exp(A^*) = (\exp A)^*$ and $\exp(TAT^{-1}) = T(\exp A)T^{-1}$. (2) If $AB = BA$, then $\exp(A+B) = (\exp A)(\exp B)$. (3) $(\exp A)(\exp(-A)) = I$. (4) There is $\varepsilon > 0$ such that exp maps the ball $B_\varepsilon = \{|A| < \varepsilon\}$ diffeomorphically onto an open neighbourhood of I.

Proof. (1) Term by term, since $(A^k)^* = (A^*)^k$, $(TAT^{-1})^k = TA^kT^{-1}$, and both operations are continuous and linear. (2) The Cauchy product of two absolutely convergent series gives $(\exp A)(\exp B) = \sum_k \sum_{i+j=k} A^i B^j/(i!\,j!) = \sum_k (A+B)^k/k!$, the binomial theorem being valid because A and B commute. (3) is (2) with $B = -A$. (4) $\exp A = I + A + R(A)$ with $|R(A)| \le \sum_{k\ge 2} |A|^k/k! \le |A|^2 e^{|A|}$, so the derivative of exp at 0 is the identity, and the inverse function theorem (as in Chapter 1 of the companion volume, *Topology from the Differentiable Viewpoint*, rewritten) applies; shrink the neighbourhood to a ball. □

We fix such an ε, taking also $\varepsilon \le 2\pi/\sqrt{n}$. The ball B_ε is invariant under $A \mapsto -A$, $A \mapsto A^*$ and $A \mapsto SAS^{-1}$ for S unitary, since these preserve $|\cdot|$. Let $\mathfrak{u}(n) = \{A : A + A^* = 0\}$ be the skew-Hermitian matrices, a real subspace of dimension n^2 (n imaginary diagonal entries and $n(n-1)/2$ complex entries above the diagonal).

Lemma 1.2. If $A \in \mathfrak{u}(n)$, then $\exp A \in U(n)$. If $A \in B_\varepsilon$ and $\exp A \in U(n)$, then $A \in \mathfrak{u}(n)$.

Proof. $(\exp A)(\exp A)^* = (\exp A)(\exp A^*) = (\exp A)(\exp(-A)) = I$ by (1) and (3). Conversely $\exp(A^*) = (\exp A)^* = (\exp A)^{-1} = \exp(-A)$, and $A^*, -A \in B_\varepsilon$, where exp is injective. □

Lemma 1.3 (Milnor's (5) and (6)). $U(n)$ is a compact smooth submanifold of $M_n(\mathbb{C})$ of dimension n^2 and a Lie group, and its tangent space at I is $\mathfrak{u}(n)$.

Proof. By Lemma 1.2, $U(n) \cap \exp(B_\varepsilon) = \exp(B_\varepsilon \cap \mathfrak{u}(n))$. In the chart $\exp^{-1}$ of $M_n(\mathbb{C})$ near I, therefore, $U(n)$ is an open piece of the linear subspace $\mathfrak{u}(n)$: near I it is a submanifold with tangent space $\mathfrak{u}(n)$. For $S_0 \in U(n)$, left multiplication $X \mapsto S_0 X$ is a linear automorphism of $M_n(\mathbb{C})$ carrying $U(n)$ onto itself and I to S_0, so it transports the chart. $U(n)$ is closed and bounded ($|S|^2 = n$), hence compact; multiplication and inversion $S \mapsto S^*$ are restrictions of polynomial maps. □

The Lie algebra. For $A \in \mathfrak{u}(n)$ the field $X_A(S) = SA$ is left invariant (left translation L_T is the linear map $X \mapsto TX$, and $T(SA) = (TS)A$), with value A at I, and every left-invariant field is of this form.

Lemma 1.4 (Milnor's "computation shows"). $[X_A, X_B] = X_{[A,B]}$, where $[A,B] = AB - BA$ and the bracket of vector fields is $[X,Y]f = X(Yf) - Y(Xf)$ (Milnor 8.5).

Proof. On an open subset of a vector space, let X, Y be given by smooth maps F, G. Then $Yf(p) = Df_p(G(p))$ and $X(Yf)(p) = D^2 f_p(F(p), G(p)) + Df_p(DG_p(F(p)))$; subtracting the same expression with F, G exchanged, the second-derivative terms cancel by symmetry, so $[X,Y]$ is given by $DG_p(F(p)) - DF_p(G(p))$. For $F(S) = SA$, $G(S) = SB$ on the open set $GL_n(\mathbb{C})$ this is $SAB - SBA$. These fields are tangent to the submanifold $U(n)$, and the bracket of fields tangent to a submanifold is tangent to it and restricts to the bracket of the restrictions. □

So the Lie algebra $\mathfrak{g}$ of $U(n)$ is $\mathfrak{u}(n)$ with the commutator.

The metric. On $\mathfrak{g}$ put $\langle A, B\rangle = \operatorname{Re}\operatorname{tr}(AB^*)$, which is positive definite: $\langle A, A\rangle = \sum |a_{ij}|^2$. Milnor defines the corresponding left-invariant metric and checks that it is right invariant by checking

invariance under the adjoint action. Both facts are contained in the following.

Lemma 1.5. Give $U(n)$ the Riemannian metric induced from the Euclidean space $M_n(\mathbb{C})$. (a) It is the left-invariant metric determined by $\langle\,,\rangle$ on $\mathfrak{g}$: $\langle SA, SB\rangle = \langle A, B\rangle$ on $TU(n)_S = S\mathfrak{g}$. (b) Left and right translations, and the inner automorphisms $X \mapsto SXS^{-1}$, are isometries. (c) The derivative of $X \mapsto SXS^{-1}$ at I is $\mathrm{Ad}(S)A = SAS^{-1}$, and $\langle \mathrm{Ad}(S)A, \mathrm{Ad}(S)B\rangle = \langle A, B\rangle$.

Proof. Multiplication on either side by a unitary matrix is a linear isometry of $M_n(\mathbb{C})$ carrying $U(n)$ onto itself, so it restricts to an isometry of $U(n)$; this gives (b), and (a) is $\operatorname{Re}\operatorname{tr}(SAB^*S^*) = \operatorname{Re}\operatorname{tr}(AB^*)$. In (c) the inner automorphism is the restriction of a linear map, which is its own derivative; and $\operatorname{tr}(SAS^{-1}(SBS^{-1})^*) = \operatorname{tr}(SAB^*S^{-1}) = \operatorname{tr}(AB^*)$, using $(S^{-1})^* = S$, exactly as on Milnor's p. 126. □

For $A, B \in \mathfrak{g}$ the trace $\operatorname{tr}(AB)$ is real, so $\langle A, B\rangle = -\operatorname{tr}(AB)$.

Geodesics. By Lemma 21.2 (Chapter 12), the geodesics through I of a Lie group with a bi-invariant metric are its one-parameter subgroups. For $A \in \mathfrak{g}$, $t \mapsto \exp(tA)$ is a one-parameter subgroup (by (2), since sA and tA commute) with velocity A at $t = 0$. Conversely, if φ is a one-parameter subgroup with $\varphi'(0) = A$, differentiating $\varphi(t+s) = \varphi(t)\varphi(s)$ at $s = 0$ gives $\varphi'(t) = \varphi(t)A$, the linear differential equation that $\exp(tA)$ also solves with the same initial value; so $\varphi(t) = \exp(tA)$. Hence *the geodesic through I with velocity A is $t \mapsto \exp(tA)$*, and the Riemannian exponential map of §10 at I is the matrix exponential (Milnor p. 125). By left translation, the geodesic through S with velocity SA is $t \mapsto S\exp(tA)$, of constant speed $|A|$.

Diagonal form; exp is onto. We quote the spectral theorem: a normal matrix is unitarily diagonalizable (e.g. Axler, *Linear Algebra Done Right*, the complex spectral theorem). A skew-Hermitian matrix has imaginary eigenvalues and a unitary one has eigenvalues of modulus one. So for $A \in \mathfrak{g}$ there is $T \in U(n)$ with $TAT^{-1} = \operatorname{diag}(ia_1, \dots, ia_n)$, and for $S \in U(n)$ there is T with $TST^{-1} = \operatorname{diag}(e^{ia_1}, \dots, e^{ia_n})$, the a_j real. In the second case $S = \exp(T^{-1}\operatorname{diag}(ia_j)T)$ by (1): $\exp : \mathfrak{g} \to U(n)$ is onto, and $U(n)$ is path connected.

The special unitary group. For $A \in \mathfrak{g}$, diagonalizing gives $\det(\exp A) = e^{\operatorname{tr} A}$. Let $SU(n) = \{S \in U(n) : \det S = 1\}$ and $\mathfrak{su}(n) = \{A \in \mathfrak{g} : \operatorname{tr} A = 0\}$, of dimension $n^2 - 1$.

Lemma 1.6. $SU(n)$ is a compact connected submanifold and Lie subgroup of $U(n)$ with tangent space $\mathfrak{su}(n)$ at I. With the induced metric it is bi-invariant, its geodesics through I are $t \mapsto \exp(tA)$, $A \in \mathfrak{su}(n)$, and $\exp : \mathfrak{su}(n) \to SU(n)$ is onto.

Proof. For $A \in B_\varepsilon \cap \mathfrak{g}$, $|\operatorname{tr} A| \le \sqrt{n}\,|A| < 2\pi$ (Cauchy–Schwarz on the diagonal), so $\exp A \in SU(n)$ iff $\operatorname{tr} A \in 2\pi i\mathbb{Z}$ iff $\operatorname{tr} A = 0$. Thus $SU(n) \cap \exp(B_\varepsilon) = \exp(B_\varepsilon \cap \mathfrak{su}(n))$, and left translation by elements of $SU(n)$ does the rest, as in Lemma 1.3. Translations by elements of $SU(n)$ are isometries of $U(n)$ preserving $SU(n)$, so the induced metric is bi-invariant, and Lemma 21.2 applies to $SU(n)$. If $S = \exp(T^{-1}\operatorname{diag}(ia_j)T) \in SU(n)$, then $\sum a_j \in 2\pi\mathbb{Z}$; changing a_1 by a multiple of 2π (which does not change S) makes $\sum a_j = 0$. □

The bracket on $\mathfrak{su}(n)$ is again the commutator. Milnor writes $\mathfrak{g}'$ for $\mathfrak{su}(n)$.

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2. Minimal geodesics from I to $-I$ (Milnor pp. 126–128)

All geodesics are parametrized by $[0, 1]$. By §1, a geodesic from I to $-I$ in $U(n)$ is $t \mapsto \exp(tA)$ with $A \in \mathfrak{g}$ and $\exp A = -I$.

Lemma 2.1. For $A \in \mathfrak{g}$, $\exp A = -I$ iff the eigenvalues of A are $i\pi k_1, \dots, i\pi k_n$ with every k_j an odd integer. The geodesic $t \mapsto \exp(tA)$, $0 \le t \le 1$, then has length $|A| = \pi(k_1^2 + \cdots + k_n^2)^{1/2}$.

Proof. With $TAT^{-1} = \operatorname{diag}(ia_j)$: $\exp A = -I$ iff $\exp(TAT^{-1}) = \operatorname{diag}(e^{ia_j}) = -I$ iff every a_j is an odd multiple of π. The speed is constantly $|A|$, and $|A| = |TAT^{-1}|$. □

Corollary 2.2. In $U(n)$ the distance from I to $-I$ is $\pi\sqrt{n}$, and the minimal geodesics from I to $-I$ are the $t \mapsto \exp(tA)$ with $A^2 = -\pi^2 I$. For such A, let P be the orthogonal projection onto $E = \text{Eigen}(i\pi) = \{v : Av = i\pi v\}$. Then $A = i\pi(2P - I)$ and

$$\exp(tA) = e^{i\pi t}P + e^{-i\pi t}(I - P). \tag{2.1}$$

Conversely, for every complex subspace $E \subseteq \mathbb{C}^n$, with projection P, the matrix $A = i\pi(2P - I)$ lies in $\mathfrak{g}$ and satisfies $A^2 = -\pi^2 I$.

Proof. $U(n)$ is compact, hence complete, so by the Hopf–Rinow theorem (Milnor 10.9) some minimal geodesic from I to $-I$ exists. By Lemma 2.1 every geodesic from I to $-I$ has length $\pi(\sum k_j^2)^{1/2} \ge \pi\sqrt{n}$, with equality iff every $k_j = \pm 1$, and equality occurs ($A = i\pi I$). So the distance is $\pi\sqrt{n}$, and the minimal geodesics are those with all $k_j = \pm 1$, i.e. with A diagonalizable with eigenvalues $\pm i\pi$, i.e. $A^2 = -\pi^2 I$. Then $\mathbb{C}^n = E \oplus E^\perp$ with $E^\perp = \text{Eigen}(-i\pi)$, which gives $A = i\pi P - i\pi(I - P)$ and (2.1). Conversely iP is skew-Hermitian because P is Hermitian, and $(2P - I)^2 = I$. □

So, as a set, the space of minimal geodesics from I to $-I$ in $U(n)$ is the set of all complex subspaces of $\mathbb{C}^n$ (Milnor p. 127). Milnor identifies it with that space; the topology needs a word. Let $\text{Gr}(\mathbb{C}^n) = \{P \in M_n(\mathbb{C}) : P = P^* = P^2\}$, a compact set, and $G_k(\mathbb{C}^n) = \{P \in \text{Gr}(\mathbb{C}^n) : \operatorname{tr} P = k\}$ (projections of rank k). It is standard that $G_k(\mathbb{C}^n)$ is a compact connected manifold of real dimension $2k(n-k)$: near a subspace E_0, the subspaces of dimension k are the graphs of the linear maps $E_0 \to E_0^\perp$.

Lemma 2.3 (the topology). Let $\Omega = \Omega(U(n); I, -I)$ carry Milnor's metric (§16)

$$d(\omega, \omega') = \max_t \rho(\omega(t), \omega'(t)) + \Big(\int_0^1 \big(\tfrac{ds}{dt} - \tfrac{ds'}{dt}\big)^2 dt\Big)^{1/2}.$$

Then $\Phi(P)(t) = e^{i\pi t}P + e^{-i\pi t}(I - P)$ is a homeomorphism of $\text{Gr}(\mathbb{C}^n)$ onto the subspace of minimal geodesics.

Proof. Φ is a bijection onto the minimal geodesics by Corollary 2.2 (P is recovered from $\Phi(P)'(0) = i\pi(2P - I)$). All the paths $\Phi(P)$ have the same constant speed $\pi\sqrt{n}$, so the integral term of $d(\Phi(P), \Phi(Q))$ vanishes. The function $(t, P, Q) \mapsto \rho(\Phi(P)(t), \Phi(Q)(t))$ is continuous on the compact set $[0,1] \times \text{Gr} \times \text{Gr}$ (the distance ρ is continuous, §10) and vanishes for $P = Q$; by uniform continuity its maximum over t tends to 0 as $Q \to P$. So Φ is continuous, and a continuous bijection from a compact space to a metric space is a homeomorphism. □

> **A hidden hypothesis.** Milnor calls the space of all subspaces "rather inconvenient", because its components $G_k(\mathbb{C}^n)$ have different dimensions. The trouble with $U(n)$ is worse: there the hypothesis of Theorem 22.1 fails. Take $k = (3, 1, \dots, 1)$. The index formula of Lemma 4.1 below holds verbatim on $\mathfrak{u}(n)$ (the extra direction iI has eigenvalue 0), and gives index $(n-1)(3-1-2) = 0$ for this non-minimal geodesic, which lies in a component of $\Omega(U(n); I, -I)$ that contains no minimal geodesic at all. Indeed, for a

path ω from I to $-I$ let θ_ω be the continuous determination of the argument of $\det \omega(t)$ with $\theta_\omega(0) = 0$; then $\theta_\omega(1)/\pi$ is an integer which cannot change along a continuous family of paths. On the geodesic $\exp(tA)$, where $\det \exp(tA) = e^{i\pi t \sum k_j}$, it equals $\sum k_j$: here $n+2$, while every minimal geodesic has $|\sum k_j| \le n$. So the hypothesis of Theorem 22.1 (every non-minimal geodesic has index $\ge \lambda_0$) fails in $U(n)$ for every $\lambda_0 \ge 1$, which is why the argument is run in $SU(2m)$: its geodesics from I to $-I$ are exactly those of $U(2m)$ with $\sum k_j = 0$.

In $SU(n)$, $\operatorname{tr} A = i\pi \sum k_j$ must vanish, and $-I \in SU(n)$ requires n even. Take $n = 2m$.

Lemma 23.1. *(checked by machine: V14.16, V14.17, restated)*[1] The space of minimal geodesics from I to $-I$ in $SU(2m)$ is homeomorphic to the complex Grassmann manifold $G_m(\mathbb{C}^{2m})$ of m-dimensional subspaces of $\mathbb{C}^{2m}$.

Proof. By Lemma 1.6 the geodesics from I to $-I$ in $SU(2m)$ are the $\exp(tA)$ with $A \in \mathfrak{su}(2m)$, $\exp A = -I$: by Lemma 2.1, eigenvalues $i\pi k_j$ with k_j odd and $\sum k_j = 0$, length $\pi(\sum k_j^2)^{1/2} \ge \pi\sqrt{2m}$. A minimal geodesic exists ($SU(2m)$ is compact), and length $\pi\sqrt{2m}$ is attained by $A = \operatorname{diag}(i\pi I_m, -i\pi I_m)$. So the minimal geodesics are those with all $k_j = \pm 1$ and $\sum k_j = 0$: m of each sign, i.e. $A = i\pi(2P - I)$ with $\operatorname{tr} P = m$. Conversely every $P \in G_m(\mathbb{C}^{2m})$ gives such an A, with $\operatorname{tr} A = i\pi(2m - 2m) = 0$. The topology is handled by Lemma 2.3, with ρ the distance of $SU(2m)$. □

Lemma 23.2. *(checked by machine: V14.11, restated; V14.20, as stated)* Every non-minimal geodesic from I to $-I$ in $SU(2m)$ has index $\ge 2m + 2$.

This is proved in §4. With Chapter 13 it gives Bott's theorem.

Theorem 23.3 (Bott). *(checked by machine: V14.22, restated)* The inclusion $G_m(\mathbb{C}^{2m}) \to \Omega(SU(2m); I, -I)$ of Lemma 23.1 induces isomorphisms of homotopy groups in dimensions $\le 2m$. Hence

$$\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m) \qquad (0 \le i \le 2m).$$

Proof. $SU(2m)$ is compact, hence complete. By Lemma 23.1 the space Ω^d of minimal geodesics is a manifold, and by Lemma 23.2 every other geodesic has index $\ge \lambda_0 = 2m + 2$. Theorem 22.1 gives $\pi_i(\Omega, \Omega^d) = 0$ for $0 \le i < 2m + 2$. In the exact homotopy sequence of the pair,

$$\pi_{i+1}(\Omega, \Omega^d) \to \pi_i(\Omega^d) \to \pi_i(\Omega) \to \pi_i(\Omega, \Omega^d),$$

both ends vanish for $i \le 2m$, so the middle map is an isomorphism (for $i = 0$ both spaces are connected: G_m is, and every component of Ω meets Ω^d). Finally, $\pi_i \Omega(M; p, q) \cong \pi_{i+1} M$ for M path connected: by Theorem 17.1, Ω is homotopy equivalent to the space of continuous paths from p to q; composing with a fixed path from q to p is a homotopy equivalence with the loop space at p; and π_i of the loop space is $\pi_{i+1} M$ (Hu, *Homotopy Theory*, p. 111, as Milnor cites; or the adjunction between suspension and loops, $\Sigma S^i = S^{i+1}$). □

[1] Checked by machine: the result has been verified in the Lean proof assistant. The numbers are those of the verified edition and of the comparison, which give the checked statement in full. *As stated*: the same statement. *Restated*: an equivalent statement in other terms. *With its hypotheses made explicit*: a hypothesis the book uses silently is stated. *In a stronger form*: the checked statement says more than the book's. *In a narrower form*: it covers less, a special setting or a weaker conclusion. *With a sign corrected*: the book's statement has a slip. *Not checked*: no part of the result has been verified. The section "Checked by machine" at the end of the chapter says which forms were checked, and covers the results that carry no such note.

For $m = 1$ this is a theorem about the round 3-sphere $SU(2)$: $G_1(\mathbb{C}^2) = \mathbb{C}P^1 = S^2$, so $\pi_1 S^2 \cong \pi_2 S^3 = 0$ and $\pi_2 S^2 \cong \pi_3 S^3 \cong \mathbb{Z}$.

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3. Stable homotopy of the unitary groups and the periodicity theorem (Milnor pp. 128–130)

Milnor now uses "standard methods of homotopy theory". These are the facts quoted.

(Q1) Coset bundles. If H is a closed subgroup of a Lie group G, then G/H is a manifold and $G \to G/H$ is a fibre bundle with fibre H; if $K \subseteq H \subseteq G$ are closed subgroups, then $G/K \to G/H$ is a fibre bundle with fibre H/K (Steenrod, *The Topology of Fibre Bundles*, §7). If G is compact and acts continuously and transitively on a Hausdorff space X with stabilizer H at x, then $gH \mapsto gx$ is a homeomorphism $G/H \to X$ (a continuous bijection from a compact space to a Hausdorff one).

(Q2) Exact homotopy sequence. For a fibre bundle $F \to E \to B$ with B path connected there is an exact sequence $\cdots \to \pi_i F \to \pi_i E \to \pi_i B \to \pi_{i-1} F \to \cdots \to \pi_0 E \to \pi_0 B$ (Steenrod §17; Hatcher, *Algebraic Topology*, Theorem 4.41 with Proposition 4.48).

(Q3) Spheres. $\pi_i S^N = 0$ for $i < N$: every map of an i-manifold into S^N with $i < N$ is homotopic to a constant, proved in Chapter 8 of the companion volume (*Topology from the Differentiable Viewpoint*, rewritten), Problem 5 (a free null-homotopy suffices, since π_1 acts on π_i fixing the zero element). $\pi_N S^N \cong \mathbb{Z}$ by the degree: Hopf's theorem, proved in Chapter 7 of the companion volume, Theorem 6.3, with Problem 4 of its Chapter 8 to pass from smooth to continuous maps; in particular $\pi_1 S^1 \cong \pi_3 S^3 \cong \mathbb{Z}$. Finally $\pi_i S^1 = 0$ for $i \geq 2$, because the covering $\mathbb{R} \to S^1$ induces isomorphisms on π_i, $i \geq 2$, and $\mathbb{R}$ is contractible (Hatcher, *Algebraic Topology*, Proposition 4.1).

The bundles needed are the following. $U(m+1)$ acts transitively on the unit sphere $S^{2m+1} \subseteq \mathbb{C}^{m+1}$ (every unit vector is the last vector of an orthonormal basis), and the stabilizer of e_{m+1} is $U(m)$, embedded as $S \mapsto \operatorname{diag}(S, 1)$; so $U(m+1)/U(m) \cong S^{2m+1}$. Embed $U(m) \subseteq U(2m)$ by $S \mapsto \operatorname{diag}(S, I_m)$; the composite of the embeddings $U(m) \subseteq U(m+1) \subseteq \cdots \subseteq U(2m)$ is this embedding. $U(2m)$ acts transitively on m-dimensional subspaces of $\mathbb{C}^{2m}$ with stabilizer $U(m) \times U(m)$ at $\mathbb{C}^m \oplus 0$; so $G_m(\mathbb{C}^{2m}) \cong U(2m)/U(m) \times U(m)$ (compare Steenrod §7). The *complex Stiefel manifold* is $V = U(2m)/U(m)$. Finally $\det : U(m) \to S^1$ is onto with kernel $SU(m)$.

Lemma 23.4. *(checked by machine: V14.25, in a stronger form; V14.26, restated; V14.27, in a narrower form)* For $1 \leq i \leq 2m$, $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i-1} U(m)$. Furthermore $\pi_{i-1} U(m) \cong \pi_{i-1} U(m+1) \cong \pi_{i-1} U(m+2) \cong \cdots$ for $i \leq 2m$, the isomorphisms being induced by the inclusions; and $\pi_j U(m) \cong \pi_j SU(m)$ for $j \neq 1$.

Proof. Stability. The bundle $U(m) \to U(m+1) \to S^{2m+1}$ gives the exact sequence

$$\pi_{j+1} S^{2m+1} \to \pi_j U(m) \to \pi_j U(m+1) \to \pi_j S^{2m+1}.$$

For $j \leq 2m-1$ both ends vanish by (Q3), so the inclusion induces an isomorphism; for $j = 2m$ the right end vanishes, so it is onto. Since $j \leq 2m-1$ implies $j \leq 2m'-1$ for every $m' \geq m$, all the later inclusions are isomorphisms too. (For $j = 0$ these are statements about connected spaces.)

The Stiefel manifold. For $i \leq 2m$, the map $\pi_i U(m) \to \pi_i U(2m)$ is onto (onto at the first step, isomorphisms afterwards, since $i \leq 2m'-1$ for $m' > m$), and $\pi_{i-1} U(m) \to \pi_{i-1} U(2m)$ is

injective ($i-1 \le 2m-1$). In the exact sequence of $U(m) \to U(2m) \to V$,

$$\pi_i U(m) \to \pi_i U(2m) \to \pi_i V \to \pi_{i-1} U(m) \to \pi_{i-1} U(2m),$$

the first map onto makes $\pi_i U(2m) \to \pi_i V$ zero, and the last map injective makes $\pi_i V \to \pi_{i-1} U(m)$ zero; by exactness $\pi_i V = 0$ for $i \le 2m$ (and V is connected).

The Grassmannian. By (Q1) with $K = U(m) \subseteq H = U(m) \times U(m) \subseteq G = U(2m)$ there is a bundle $U(m) \to V \to G_m(\mathbb{C}^{2m})$. Its sequence $\pi_i V \to \pi_i G_m \to \pi_{i-1} U(m) \to \pi_{i-1} V$ has vanishing ends for $1 \le i \le 2m$.

SU versus U. The bundle $SU(m) \to U(m) \to S^1$ gives $\pi_{j+1} S^1 \to \pi_j SU(m) \to \pi_j U(m) \to \pi_j S^1$, with vanishing ends for $j \ge 2$; for $j = 0$ both groups are connected. □

Definition. For $j \ge 0$ the mutually isomorphic groups $\pi_j U(m)$, $2m - 1 \ge j$, are the j-th *stable homotopy group of the unitary group*, written $\pi_j U$. The proof also showed (Milnor p. 129) that $\pi_{2m} U(m) \to \pi_{2m} U(m+1) \cong \pi_{2m} U$ is onto.

Periodicity theorem. $\pi_{i-1} U \cong \pi_{i+1} U$ for all $i \ge 1$.

Proof. Given $i \ge 1$, choose m with $2m \ge i$. Then

$$\pi_{i-1} U = \pi_{i-1} U(m) \cong \pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m) \cong \pi_{i+1} U(2m) = \pi_{i+1} U,$$

by Lemma 23.4, Theorem 23.3, Lemma 23.4 again ($i+1 \ge 2$), and stability ($i+1 \le 2m+1 \le 2(2m)-1$). □

Theorem 23.5 (Bott). *(checked by machine: V14.29, V14.30, restated)* The stable homotopy groups $\pi_i U$ are periodic of period 2: $\pi_0 U \cong \pi_2 U \cong \pi_4 U \cong \cdots$ are zero, and $\pi_1 U \cong \pi_3 U \cong \pi_5 U \cong \cdots$ are infinite cyclic.

Proof. $U(1) = S^1$ and $0, 1 \le 2 \cdot 1 - 1$, so $\pi_0 U = \pi_0 S^1 = 0$ and $\pi_1 U = \pi_1 S^1 \cong \mathbb{Z}$; periodicity does the rest. As a check, independent of periodicity: $SU(2)$ is the 3-sphere of unit quaternions, and $2, 3 \le 2 \cdot 2 - 1$, so by Lemma 23.4 $\pi_2 U = \pi_2 SU(2) = 0$ and $\pi_3 U = \pi_3 SU(2) \cong \mathbb{Z}$. □

Milnor's isomorphisms are abstract: they are composites of isomorphisms chosen for each i separately. Bott stated his theorem as a homotopy equivalence of spaces. That form also follows from Theorem 23.3, once one passes to the limit $m \to \infty$ with some care; it is not in Milnor's text.

Corollary 23.6 (the space form of Bott's theorem). Let $BU = \bigcup_m G_m(\mathbb{C}^{2m})$, along $E \mapsto E \oplus \mathbb{C}e_{2m+1}$, let $SU = \bigcup_m SU(2m)$ and $U = \bigcup_m U(2m)$, along $S \mapsto \operatorname{diag}(S, I_2)$, each with the colimit topology (a set is closed iff it meets every stage in a closed set), and let ΩX be the space of continuous loops at the identity with the compact-open topology. Then there is a map $\beta : BU \to \Omega SU$ inducing isomorphisms of all homotopy groups, and a map $\mathbb{Z} \times BU \to \Omega U$ with the same property.

Proof. Let $\varphi_m(E)(t) = e^{i\pi t} P_E + e^{-i\pi t}(I - P_E)$ be the minimal geodesic (2.1), let $E_0 = \operatorname{span}(e_1, e_3, \dots, e_{2m-1})$, and put $\beta_m(E)(t) = \varphi_m(E)(t)\, \varphi_m(E_0)(t)^{-1}$, a loop at I in $SU(2m)$. Pointwise multiplication by the fixed path $\varphi_m(E_0)^{-1}$ is a homeomorphism from the paths from I to $-I$ onto the loops at I (its inverse multiplies by $\varphi_m(E_0)$), so by Theorems 23.3 and 17.1, β_m induces isomorphisms on π_i for $i \le 2m$. Since $\varphi_{m+1}(E \oplus \mathbb{C}e_{2m+1}) = \operatorname{diag}(\varphi_m(E), e^{i\pi t}, e^{-i\pi t})$ and E_0 was chosen to be stable in the same way, $\beta_{m+1}(E \oplus \mathbb{C}e_{2m+1}) = \operatorname{diag}(\beta_m(E), I_2)$: the β_m fit together to a map β.

Compact sets in a colimit. Let $X_0 \subseteq X_1 \subseteq \cdots$ be T_1 spaces, each a closed subspace of the next, and $X = \bigcup X_m$ with the colimit topology. Then each X_m is a closed subspace of X (a set closed in X_m meets each X_j in a closed set), and every compact $K \subseteq X$ lies in some X_m: otherwise pick $x_1, x_2, \ldots \in K$ and $m_1 < m_2 < \cdots$ with $x_k \in X_{m_k} \setminus X_{m_{k-1}}$; every subset of $\{x_k\}$ meets each X_j in a finite, hence closed, set, so $\{x_k\}$ is an infinite closed discrete subset of the compact K, which is impossible.

Consequently a map $S^i \to BU$, or a homotopy $S^i \times [0,1] \to BU$, factors through some stage, and $\pi_i BU = \operatorname{colim}_m \pi_i G_m(\mathbb{C}^{2m})$. A map $S^i \to \Omega SU$ is the same as a map $S^i \times S^1 \to SU$ (exponential law; S^1 is compact Hausdorff, Hatcher, Appendix, on the compact-open topology), whose compact image lies in some $SU(2m)$; so likewise $\pi_i \Omega SU = \operatorname{colim}_m \pi_i \Omega SU(2m)$. In each degree i the maps β_{m*} are isomorphisms for $m \geq i/2$, so β_* is an isomorphism.

For U: $S \mapsto (S\, D(\det S)^{-1}, \det S)$, with $D(z) = \operatorname{diag}(z, 1, \ldots, 1)$, is a homeomorphism $U(2m) \cong SU(2m) \times S^1$ compatible with the inclusions. Define $\mathbb{Z} \times BU \to \Omega U$ by $(N, E) \mapsto D(e^{2\pi i N t})\, \beta(E)(t)$. The winding number of the determinant of this loop is N, so the map is a bijection on components; on the component N it is β followed by the inclusion $\Omega SU \to \Omega_0 U$ and by the homeomorphism "multiply by $D(e^{2\pi i N t})$". The inclusion $\Omega SU \to \Omega_0 U$ induces isomorphisms on all π_i, since $\pi_i \Omega SU = \pi_{i+1} SU \cong \pi_{i+1} U = \pi_i \Omega_0 U$ for $i \geq 1$ (Lemma 23.4, passed to the colimit), and both spaces are connected ($\pi_1 SU(2m) \cong \pi_0 G_m(\mathbb{C}^{2m}) = 0$ by Theorem 23.3). □

> **Milnor's text skips the passage to spaces.** To upgrade Corollary 23.6 to homotopy equivalences $\Omega SU \simeq BU$ and $\Omega U \simeq \mathbb{Z} \times BU$ one needs that the spaces involved have the homotopy type of CW complexes, and then Whitehead's theorem (Hatcher, Theorem 4.5). We quote this. The colimits are taken along smooth embeddings of compact manifolds, which are closed cofibrations (tubular neighbourhoods), so each colimit is homotopy equivalent to the mapping telescope of compact manifolds, a space of CW homotopy type; and the loop space of a space of CW homotopy type has CW homotopy type (J. Milnor, *On spaces having the homotopy type of a CW-complex*, Trans. AMS 90 (1959)). Milnor appeals to the same two facts, without proof, in the proof of Theorem 24.5 (p. 146). For the homotopy groups, which is all that Theorem 23.5 asserts, they are not needed.

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4. The index of a non-minimal geodesic (Milnor pp. 130–132)

We now prove Lemma 23.2. The tools are Theorem 20.5 (Chapter 12): if M is locally symmetric and γ is a geodesic with $\gamma(0) = p$ and $\gamma'(0) = V$, let $K_V(W) = R(V, W)V$ on TM_p; then the parameters $t \neq 0$ conjugate to 0 along γ are the $t = \pi s/\sqrt{e}$, s a non-zero integer and e a positive eigenvalue of K_V, and the multiplicity of t is the number of eigenvalues e, counted with multiplicity, such that t is a multiple of $\pi/\sqrt{e}$. The Index Theorem 15.1 (Chapter 9): the index of a geodesic $\gamma : [0,1] \to M$ is the number of parameters $t \in (0,1)$ such that $\gamma(t)$ is conjugate to $\gamma(0)$ along γ, counted with multiplicity. And Theorem 21.3(b): for left-invariant fields on a Lie group with bi-invariant metric, $R(X,Y)Z = \frac{1}{4}[[X,Y],Z]$.

Lemma 4.1 (the index formula). Let n be even and $A \in \mathfrak{su}(n)$ with eigenvalues $i\pi k_1, \ldots, i\pi k_n$, the k_j odd with $\sum k_j = 0$. The index of the geodesic $t \mapsto \exp(tA)$, $0 \leq t \leq 1$, from I to $-I$ in

$SU(n)$ is

$$\lambda = \sum_{k_j > k_l} (k_j - k_l - 2),$$

the sum over all pairs of indices $\{j, l\}$ with $k_j \neq k_l$, written with the larger first.

Proof. Reduction to diagonal form (Milnor's "we may assume"). For $T \in U(n)$, conjugation $S \mapsto TST^{-1}$ is an isometry of $SU(n)$ (Lemma 1.5) fixing I and $-I$ and carrying $\exp(tA)$ to $\exp(tTAT^{-1})$. An isometry carries Jacobi fields to Jacobi fields, so it preserves conjugate points and their multiplicities, hence (Theorem 15.1) indices. Choose T, including a permutation, so that $A = \operatorname{diag}(i\pi k_1, \dots, i\pi k_n)$ with $k_1 \geq \cdots \geq k_n$.

The operator K_A. $SU(n)$ with its bi-invariant metric is a symmetric space (Lemma 21.1), so Theorem 20.5 applies with $p = I$, $V = A$. By Theorem 21.3(b) and Lemma 1.4,

$$K_A(W) = R(A, W)A = \tfrac{1}{4}[[A, W], A] = -\tfrac{1}{4}[A, [A, W]], \qquad W \in \mathfrak{su}(n).$$

For $W = (w_{jl})$, $[A, W]_{jl} = i\pi k_j w_{jl} - w_{jl}\, i\pi k_l = i\pi(k_j - k_l)w_{jl}$, hence $[A, [A, W]]_{jl} = -\pi^2(k_j - k_l)^2 w_{jl}$ and

$$K_A(W) = \Big(\tfrac{\pi^2}{4}(k_j - k_l)^2\, w_{jl}\Big).$$

An eigenbasis. For $j < l$ let E_{jl} have $+1$ in place (j, l), -1 in place (l, j) and zeros elsewhere, and let E'_{jl} have i in both places. Both lie in $\mathfrak{su}(n)$ and are eigenvectors with eigenvalue $\frac{\pi^2}{4}(k_j - k_l)^2$. Every diagonal matrix in $\mathfrak{su}(n)$ is an eigenvector with eigenvalue 0. These are $n(n-1)+(n-1) = n^2 - 1 = \dim \mathfrak{su}(n)$ independent vectors, so they form a basis. The positive eigenvalues of K_A are therefore the numbers $e_{jl} = \frac{\pi^2}{4}(k_j - k_l)^2$ for the pairs with $k_j > k_l$, each counted twice.

Conjugate points. For $e = e_{jl}$, $\pi/\sqrt{e} = 2/(k_j - k_l)$, so this eigenvalue produces conjugate points at $t = 2/(k_j - k_l), 4/(k_j - k_l), \dots$. Since $k_j - k_l$ is even, the number of these in the open interval $(0, 1)$ is $(k_j - k_l)/2 - 1$.

The index. By Theorems 20.5 and 15.1, λ is the sum over the positive eigenvalues e (with multiplicity) of the number of multiples of $\pi/\sqrt{e}$ in $(0, 1)$. Each pair $k_j > k_l$ contributes $2((k_j - k_l)/2 - 1) = k_j - k_l - 2$. $\square$

For a minimal geodesic all $k_j = \pm 1$, every term is $1 - (-1) - 2 = 0$, and $\lambda = 0$, as it should be. Every term of the sum is ≥ 0, so any set of pairs gives a lower bound.

Proof of Lemma 23.2. Let $n = 2m$, $\sum k_j = 0$, not all $k_j = \pm 1$. Let P and N be the numbers of positive and of negative k_j, so $P + N = 2m$.

Case 1: $N \geq m + 1$ (Milnor's "at least $m + 1$ of the k_i are (say) negative"). Then $P \leq m - 1$, while the positive k_j add up to $\sum_{k_l < 0} |k_l| \geq N \geq m + 1 > P$; so some $k_{j_0} \geq 3$. Pairing k_{j_0} with each of the N negative entries, each term is $\geq 3 + 1 - 2 = 2$, and $\lambda \geq 2N \geq 2(m + 1)$. The case $P \geq m + 1$ is the same with all signs reversed: the formula is unchanged under $k \mapsto -k$.

Case 2: $P = N = m$. Some $|k_j| \geq 3$; say $k_{j_0} \geq 3$, the other sign being symmetric. The positive entries then add up to at least $(m - 1) + 3 = m + 2 > m = N$, and they equal $\sum_{k_l < 0} |k_l|$, so some $k_{l_0} \leq -3$. Take the pairs (j_0, l) for the $m - 1$ negative $l \neq l_0$, the pairs (j, l_0) for the $m - 1$ positive $j \neq j_0$, and the pair (j_0, l_0):

$$\lambda \geq (m - 1)(3 + 1 - 2) + (m - 1)(1 + 3 - 2) + (3 + 3 - 2) = 4m \geq 2(m + 1).$$

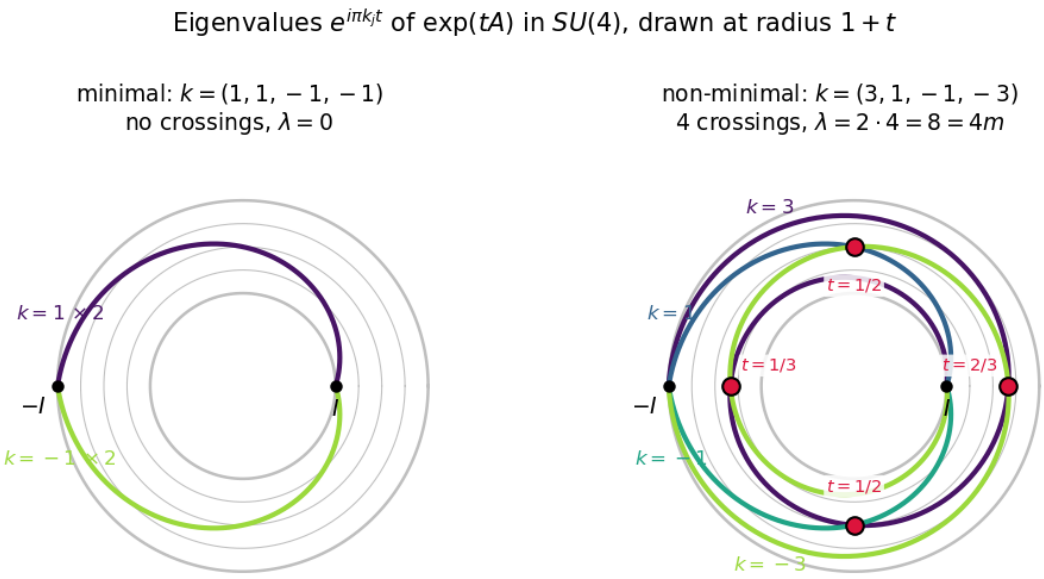
In either case $\lambda \geq 2m+2$. $\square$

This completes the proof of Theorem 23.3, and with it of Theorem 23.5. Both bounds are attained: for $m=2$, $k=(3,-1,-1,-1)$ gives $\lambda = 6 = 2m+2$ and $k=(3,1,-1,-3)$ gives $\lambda = 8 = 4m$.

An interactive look at the eigenvalue braid

Along the geodesic $t \mapsto \exp(tA)$ the eigenvalues of $\exp(tA)$ are $e^{i\pi k_j t}$: they leave 1 together at $t=0$, run around the unit circle at speeds k_j, and all arrive at -1 at $t=1$. Two of them, with $k_j \neq k_l$, coincide exactly when t is a multiple of $2/(k_j-k_l)$, which is where the proof of Lemma 4.1 finds the conjugate points coming from the pair (j,l). So the index is twice the number of collisions strictly between $t=0$ and $t=1$. The figure draws the eigenvalue at time t at radius $1+t$, so that each eigenvalue traces a spiral from the inner circle to the outer one and each collision is a crossing of two spirals. Drag the time slider to move the eigenvalues along; choose m; load a minimal geodesic (no crossings: half the eigenvalues turn one way, half the other), a random non-minimal one, or "pump" the extreme speeds apart ($k_1 \mapsto k_1+2$, $k_{2m} \mapsto k_{2m}-2$). The panel shows the collisions passed so far, the index from the formula, and Milnor's bound $2m+2$.

(In the printed version this figure is a still image; the interactive version is in the web edition.)



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5. Application: complex vector bundles over spheres

Clutching. A complex vector bundle of rank N over S^n is trivial over each closed hemisphere (a contractible space), so it is obtained from two trivial bundles by gluing along the equator S^{n-1} with a map $f : S^{n-1} \to GL_N(\mathbb{C})$, and isomorphism classes of bundles correspond to homotopy classes of such maps (Steenrod §18; Hatcher, *Vector Bundles and K-Theory*, Proposition 1.11). Gram–Schmidt deforms $GL_N(\mathbb{C})$ onto $U(N)$, and since $U(N)$ is a connected topological group, free

homotopy classes of maps $S^{n-1} \to U(N)$ are the same as elements of $\pi_{n-1}U(N)$. So

$$\{\text{rank } N \text{ complex bundles over } S^n\}/\cong \;=\; \pi_{n-1}U(N).$$

By Lemma 23.4 this is the stable group $\pi_{n-1}U$ as soon as $n-1 \le 2N-1$, i.e. $N \ge n/2$, and by Theorem 23.5:

sphere	S^1	S^2	S^3	S^4	S^5	S^6	S^7	S^8
$\pi_{n-1}U$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
stable for	$N \ge 1$	$N \ge 1$	$N \ge 2$	$N \ge 2$	$N \ge 3$	$N \ge 3$	$N \ge 4$	$N \ge 4$

S^2: monopoles. $\pi_1 U(1) = \mathbb{Z}$, and the clutching map of the bundle with integer N is $z \mapsto z^N$ on the equator. This is Dirac's integer. The tautological line bundle of $\mathbb{C}P^1 = S^2$ has $|N| = 1$; the tangent bundle of S^2, a complex line bundle, has clutching map $z \mapsto z^{\pm 2}$, so $|N| = 2$, and this 2 is the Euler number $\chi(S^2)$ (Chapter 6 of the companion volume, where the Poincaré–Hopf theorem is proved). By stability, every complex bundle over S^2 of any rank is a line bundle plus a trivial bundle.

S^4: instantons. $\pi_3 U(2) = \pi_3 SU(2) = \mathbb{Z}$, already stable. The clutching map $S^3 \to SU(2)$ that identifies the unit quaternions with $SU(2)$ has degree one; the resulting rank-2 bundle is the quaternionic Hopf bundle over $\mathbb{H}P^1 = S^4$, and the integer is the instanton number of Yang–Mills theory (the BPST instanton has instanton number 1).

Odd spheres. Since $\pi_{2k}U = 0$, every complex bundle of rank $\ge k+1$ over S^{2k+1} is trivial, and every complex bundle over S^1 or S^3 is trivial.

Below the stable range the answer is irregular, which is why the theorem is about U and not about a fixed $U(N)$. Since $U(2) \cong SU(2) \times S^1$ as spaces and $SU(2) = S^3$, rank-2 bundles over S^n are classified by $\pi_{n-1}S^3$ for $n \ge 3$, and the standard values $\pi_4 S^3 = \mathbb{Z}/2$, $\pi_5 S^3 = \mathbb{Z}/2$, $\pi_6 S^3 = \mathbb{Z}/12$ give: over S^5 there is exactly one non-trivial rank-2 bundle, and it becomes trivial when a trivial line bundle is added (rank 3 is stable, and $\pi_4 U = 0$); over S^6 the rank-2 bundles form $\mathbb{Z}/2$ but those of rank ≥ 3 form $\mathbb{Z}$; over S^7 the rank-2 bundles form $\mathbb{Z}/12$, those of rank ≥ 4 are all trivial.

In K-theory language. The reduced K-group $\widetilde{K}(S^n)$ of stable bundles is $\pi_{n-1}U$, and Theorem 23.5 reads $\widetilde{K}(S^{n+2}) \cong \widetilde{K}(S^n)$. The integer attached to a stable bundle over S^2 is its first Chern number, and the same integer, for the bundle of occupied electron states over the two-dimensional Brillouin torus, is the quantized Hall conductance in units of e^2/h (Thouless, Kohmoto, Nightingale and den Nijs, 1982). In Kitaev's classification of topological insulators (2009), insulators without symmetries in d dimensions carry an integer invariant of this kind exactly when $\widetilde{K}(S^d) = \pi_{d-1}U$ is $\mathbb{Z}$, that is, for d even: the period two of that column of Kitaev's table is the period two of Theorem 23.5.

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6. Summary of what was silently assumed, now made explicit

Where	Milnor says	What is actually needed
p. 124	properties (1)–(4) "are easily verified"	absolute convergence; the Cauchy product with commuting factors; the inverse function theorem (Lemma 1.1)
p. 124	"one easily proves" that $U(n)$ is a submanifold with $TU(n)_I$ skew-Hermitian	exp on a small ball is a chart in which $U(n)$ is a linear subspace; left translation elsewhere (Lemmas 1.2, 1.3)
p. 124	"Computation shows" the bracket is $AB - BA$	bracket of linear vector fields on $GL_n(\mathbb{C})$, restricted to a submanifold (Lemma 1.4)
p. 125	matrix exp equals the Riemannian exp	Lemma 21.2, plus uniqueness for $\varphi' = \varphi A$
p. 125–126	right invariance via Ad	equivalently, the metric is induced from $M_n(\mathbb{C})$, on which unitary multiplications are isometries (Lemma 1.5)
p. 126	"by ordinary matrix theory" A can be diagonalized	spectral theorem for normal matrices (quoted)
p. 126	the Lie algebra of $SU(n)$ is the trace-zero part	$\det \exp A = e^{\operatorname{tr} A}$ and the bound $\lvert\operatorname{tr} A\rvert < 2\pi$ on a small ball (Lemma 1.6)
p. 127	minimal geodesics "may be identified" with all subspaces	existence of a minimal geodesic (Hopf–Rinow 10.9) and a homeomorphism, not just a bijection (Lemma 2.3)
p. 127	$U(n)$ is "rather inconvenient"	worse than inconvenient: $k = (3, 1, \dots, 1)$ is a non-minimal geodesic of index 0, so the hypothesis of Theorem 22.1 fails in $U(n)$ (§2)
p. 127	for $SU(2m)$ "all the above considerations remain valid"	the length bound and existence of a minimal geodesic inside $SU(2m)$; $-I \in SU(n)$ needs n even
p. 128	Theorem 23.3 from §22	exact sequence of the pair; $\pi_i\Omega(M; p, q) \cong \pi_{i+1}M$ via Theorem 17.1
pp. 128–129	the fibrations and their exact sequences	coset spaces of Lie groups are bundles and have exact sequences (quoted, (Q1), (Q2)); $\pi_i S^N = 0$ for $i < N$ and $\pi_N S^N = \mathbb{Z}$ (companion volume, (Q3))
p. 129	$\pi_i(U(2m)/U(m)) = 0$ for $i \le 2m$	$\pi_i U(m) \to \pi_i U(2m)$ onto for $i \le 2m$, injective for $i \le 2m - 1$
p. 129	"Thus we obtain" the periodicity theorem $\pi_{i-1}U \cong \pi_{i+1}U$	an abstract isomorphism for each i; the space-level statement needs colimits (Corollary 23.6) and CW homotopy types (quoted)
p. 130	"a short computation shows" $[A, W]$	entrywise formula and the eigenbasis E_{jl}, E'_{jl}, diagonal (Lemma 4.1)
p. 130	"We may assume that A is the diagonal matrix"	conjugation by a unitary matrix is an isometry of $SU(n)$ fixing $\pm I$, so it preserves indices

Where	Milnor says	What is actually needed
p. 131	the count is "evidently" $(k_j - k_l)/2 - 1$	$k_j - k_l$ is even
p. 132	Case 1, "(say) negative", "one of the positive k_i must be ≥ 3"	the formula is invariant under $k \mapsto -k$; the positive entries add up to more than their number
p. 132	Case 2, "one is ≥ 3 and one is ≤ -3"	$\sum k_j = 0$ and $P = N = m$

Checked by machine. Milnor's §23 has been verified in the Lean proof assistant. The algebra and counting of §§2 and 4: for $A = \operatorname{diag}(i\pi k_1, \dots, i\pi k_n)$, $\exp A = -I$ iff every k_j is odd; $|A|^2 = \pi^2 \sum k_j^2 \geq \pi^2 n$, with equality iff all $k_j = \pm 1$, and then m of each sign in $SU(2m)$; Milnor's formula for K_A, with eigenvectors shown to span; and, from this spectrum, the index formula of Lemma 4.1, Lemma 23.2, and the non-minimal geodesic $k = (3, 1, \dots, 1)$ of index 0 in $U(n)$. For every skew-Hermitian A, $A^2 = -\pi^2 I$ iff $A = i\pi(2P - I)$ with P a Hermitian projection, the algebraic content of Corollary 2.2 and Lemma 23.1. The geometry and the topology: $U(n)$ and $SU(n)$ as compact submanifolds of $M_n(\mathbb{C})$ with the induced bi-invariant metric; that every geodesic from I to $-I$ in $SU(n)$ is $t \mapsto \exp(tA)$ with A unitarily conjugate to the diagonal form; the distance from I to $-I$ in $SU(2m)$ and its minimal geodesics; Lemma 23.1, as a homeomorphism for the initial velocities and, in the path space, as a homotopy equivalence (a homeomorphism after identifying paths that agree on $[0, 1]$); Theorem 20.5 for $SU(n)$, so that the index formula of Lemma 4.1 and Lemma 23.2 hold for the geodesics themselves, by the Index Theorem; an explicit neighbourhood retraction, which makes Theorem 23.3 independent of Hanner's theorem; Theorem 23.3; the bundles of Lemma 23.4 on concrete Stiefel manifolds and Grassmannians, with their exact homotopy sequences, and Lemma 23.4; the periodicity theorem, inside one $U(m)$ for $1 \leq i \leq 2m - 2$ and between any two groups in the stable range; and Theorem 23.5. Each isomorphism is a bijection of pointed sets and, in the range of one $U(m)$ and where both sides are groups, an isomorphism of groups. Not checked: $U(n)$ beyond what $SU(2m)$ needs (its minimal geodesics, the connectedness of $U(n)$ and $SU(n)$, Lemma 2.3 for $U(n)$), the stable groups $\pi_i U$ as homotopy groups of a limit, Corollary 23.6, and the application. See the verified edition.